

VIVAAN THADANI

289-716-9122 | vivaanthadani@gmail.com | linkedin.com/in/vivaanthadani | Portfolio: vivaanthadani.com

SUMMARY OF QUALIFICATIONS

- Hands-on mechanical engineering experience across 3+ major technical projects, applying 5+ design, analysis, manufacturing, and testing methods including CAD, FEA, 3D printing, CNC machining, and physical testing.
- Demonstrated real-world engineering problem solving through field-tested equipment improvements, including a solar-powered camera system that reduced maintenance costs by \$6,000 annually per site.
- Proven communication, adaptability, and time-management skills developed over 3 summers in a fast-paced camp environment, balancing the needs of 20–30 campers while clearly communicating behavioural and safety concerns.
- Demonstrated project leadership and multidisciplinary collaboration through Hyperloop subsystem development and a 7-member mechanical design team progressing from concept through fabrication and iterative testing.

SKILLS

- **Engineering Software:** SolidWorks (CSWA), ANSYS, Fusion 360, AutoCAD, Bambu Studio
- **Core Engineering Skills:** CAD Modelling, GD&T, Mechanical Drawings, Structural Analysis, 3D Printing
- **Programming and Data Tools:** MATLAB, Simulink, Python, C, Arduino, Microsoft Excel

EDUCATION

Toronto Metropolitan University

Sep 2024 – April 2029 (Expected)

Bachelor of Engineering, Mechanical Engineering, cGPA: 3.83/4.33

Dean's List: 2024, 2025

- **Relevant Coursework:** Manufacturing, Engineering Design, Mechanics of Materials, Circuits, Dynamics, Thermodynamics, Fluid Mechanics, Materials Science, Electric Machines & Actuators

EXPERIENCE + PROJECTS

Engineering Technical Trainee – Toronto Water

May 2026 – Aug 2026

- Designed and created a solar powered wiper for a live feed sewer camera using Fusion 360, Arduino and 3D printing for the sewer environment, resulting in a maintenance cost reduction of \$6000 annually per camera site.
- Engineered and validated custom TPU wheels for a sewer inspection crawler through dimensional measurement, CAD modelling, material selection, 3D printing and field testing to improve traction, stability, and serviceability.
- Reverse-engineered a sewer flushing nozzle through component disassembly, datasheet analysis, and wiring diagrams; researched and selected upgraded hardware for an integrated live-feed HD camera system.
- Developed and validated a sensor inventory application through edge-case testing and issue tracking; created user guides and workflow documentation and trained 40+ operators on asset tracking and system use.

Structures Project Lead – TMU Hyperloop

Oct 2025 – Present

- Lead R&D of a PCM-assisted water–glycol cooling system for LIMs by validating materials through literature review and coordinating piping/enclosure design, BOM development, and weekly design reviews.
- Performed structural FEA in ANSYS Mechanical on LIM mounts and 2024 aluminum brackets by calculating loads, defining materials, meshes, and boundary conditions, and documenting stress/deformation results.
- Coordinated mechanical and electrical interfaces across vehicle systems to improve subsystem compatibility and support cross-functional integration.

Gear-Driven Autonomous Wind-Up Car – MEC322 Design Project

Jan 2026 – Apr 2026

- Collaborated in a cross-functional team of seven to deliver a fully functional mechanical prototype from concept to competition, managing design integration, manufacturing constraints, and iterative testing within a structured engineering design process.
- Performed engineering calculations for gear geometry, torque transmission, drive force, and travel distance to validate a 3:1 cycloid gear reduction within strict material and dimensional constraints.
- Developed a complete SolidWorks CAD drawing package with assembly and detail drawings and motion simulation, then fabricated and iteratively tested a laser-cut acrylic prototype.